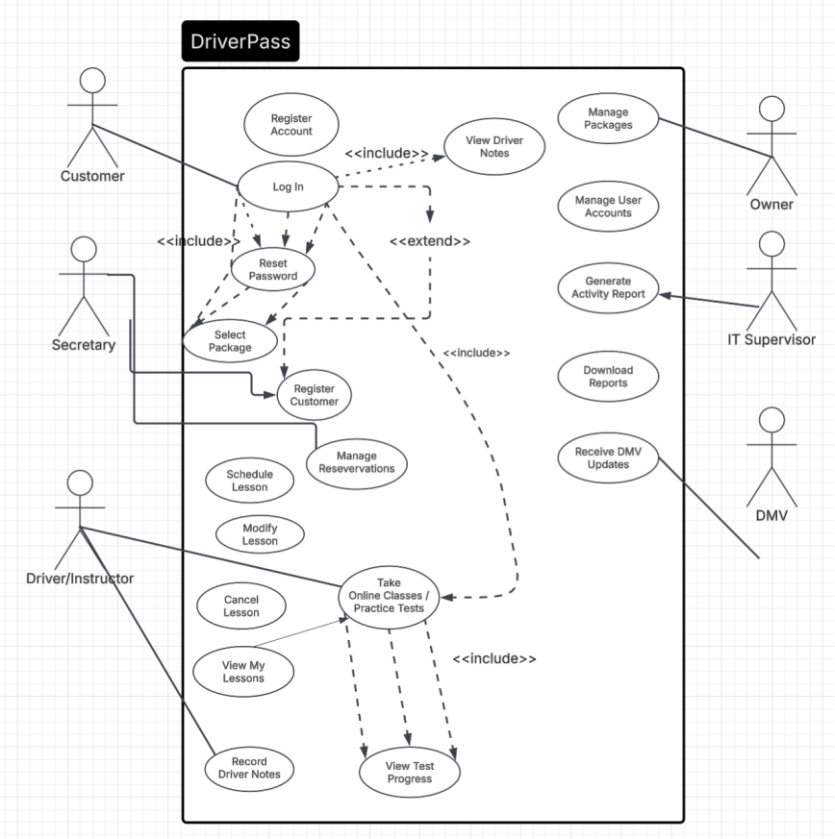
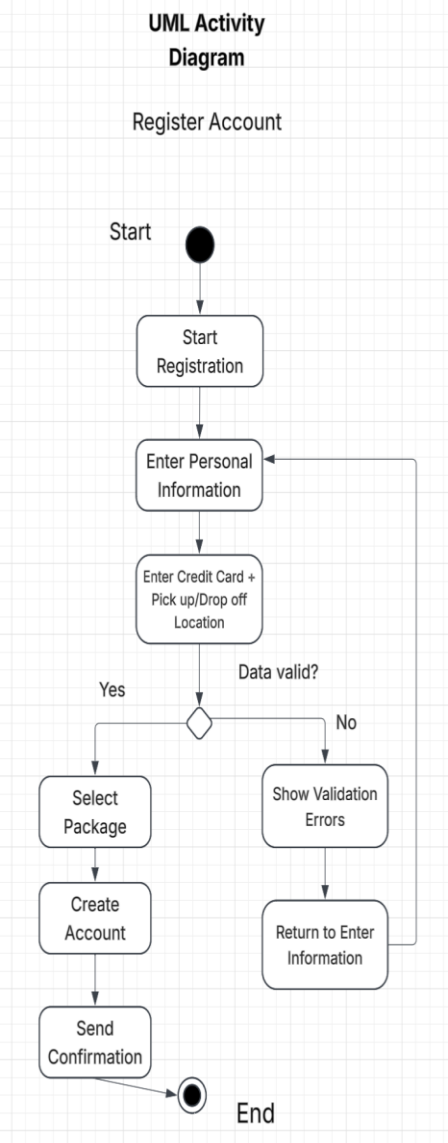
CS 255 System Design Document Template

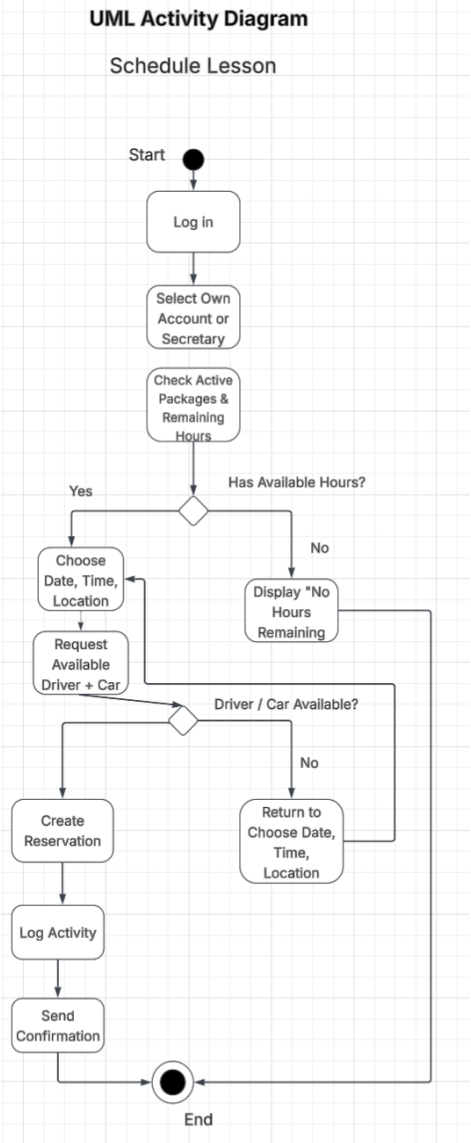
UML Diagrams

UML Use Case Diagram

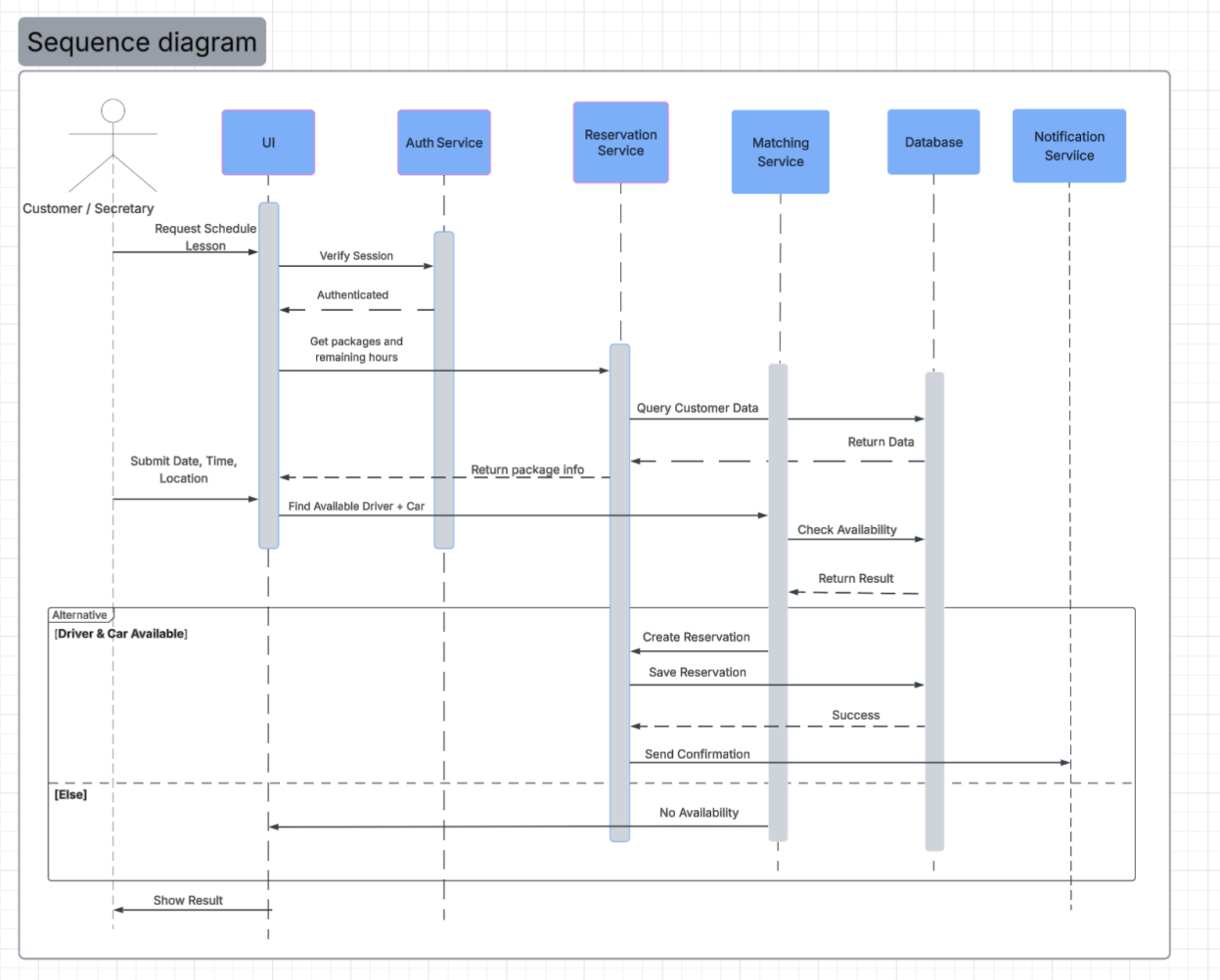


UML Activity Diagrams

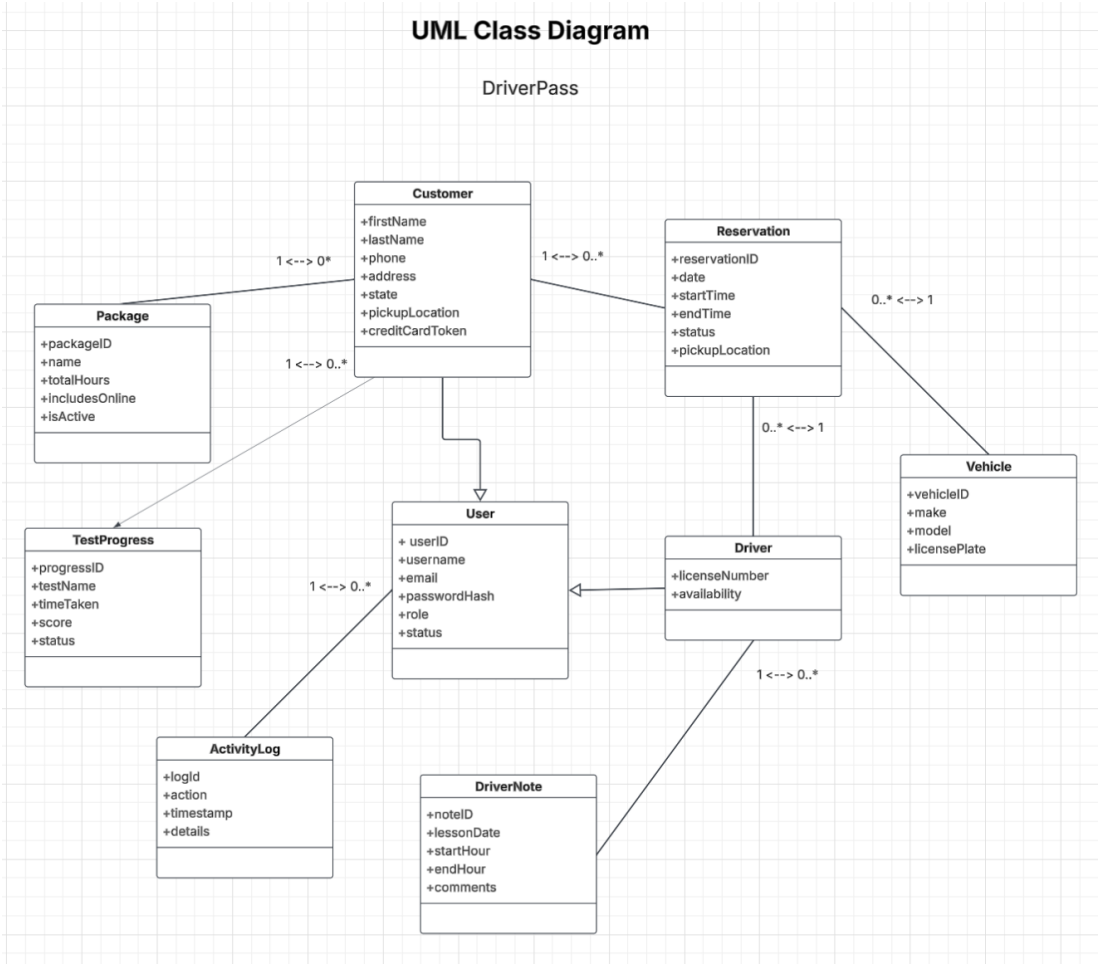




UML Sequence Diagram



UML Class Diagram



Technical Requirements

The DriverPass system is designed as a cloud-based web application so that DriverPass does not need to manage its own security or servers.

Hardware

- End users such as customers, staff, and drivers only need a computer, tablet, or smartphone with an internet connection and a modern web browser.
- No special hardware is required on the client side.

Software

- The system will run in standard web browsers such as Chrome, Safari, Edge, and Firefox on both desktop and mobile devices.
- A relational database is required to store customer information, reservations, packages, lesson notes, activity logs, and test progress.

Tools and Infrastructure

- Secure HTTPS/TLS connection for all data transferred between users and the system.
- Cloud hosting platform such as AWS, Azure, or similar, to host the application and database.
- Standard development and CASE tools (LucidChart for diagrams) were used during the design phase.
- Future implementation will use common web technologies (front-end framework + backend API + relational database).

These technical requirements support the functional needs of the system while keeping infrastructure management simple for DriverPass.